MIL-C-27652D(USAF)
4 December 1980
SUPERSEDING
MIL-C-27652C(USAF)
18 November 1963

MILITARY SPECIFICATION

CONVERTER, LIQUID OXYGEN GCU-20/A

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of converter. The converter is designated GCU-20/A. The converter does not have integral warming coils.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-V-9050	Valves, Oxygen, Pressure Relief, Aircraft
MIL-C-25516	Connectors, Electrical, Miniature, Coaxial, Environment Resistant Type, General Specification for
MIL-C-25666	Converters, Liquid Oxygen, Capacitance Type Gaging, General Specification for

STANDARD

MILITARY

MIL-STD-831	Test Reports, Preparation of

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurements should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General specification. Unless otherwise specified herein, the converter shall conform to the requirements specified in MIL-C-25666.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENESS, Wright-Patterson AFB, OH 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1660

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3.2 Qualification. Converters furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.3 Design and construction. The converter shall conform to figures 1 through 3. The converter shall have an operating pressure of 300 pounds per square inch gage (psig) and a minimum capacity of 75 liters of liquid oxygen measured at 1 atmosphere plus at least 3 percent of this volume to allow for the change in liquid density at the operating pressure.

3.3.1 Strength. The filled converter shall be designed to withstand the dynamic loads that are imposed by the following accelerations:

Vertical down - 9.0 G
Vertical up - 3.0 G
Lateral - 4.0 G.

3.3.2 Marking of ports. The ports shall be marked as indicated in figure 2.

3.4 Performance

3.4.1 Final assembly pressure. The final assembly pressure of the converter shall be 320 psig.

3.4.2 Filling time. The time required to fill the converter shall not exceed 30 minutes.

3.4.3 Evaporation loss

3.4.3.1 Vented condition. The evaporation loss of the essentially full converter in the vent condition shall not exceed 9 pounds.

3.4.3.2 Buildup condition. The evaporation loss of the essentially full converter in the buildup condition shall not exceed 5 pounds.

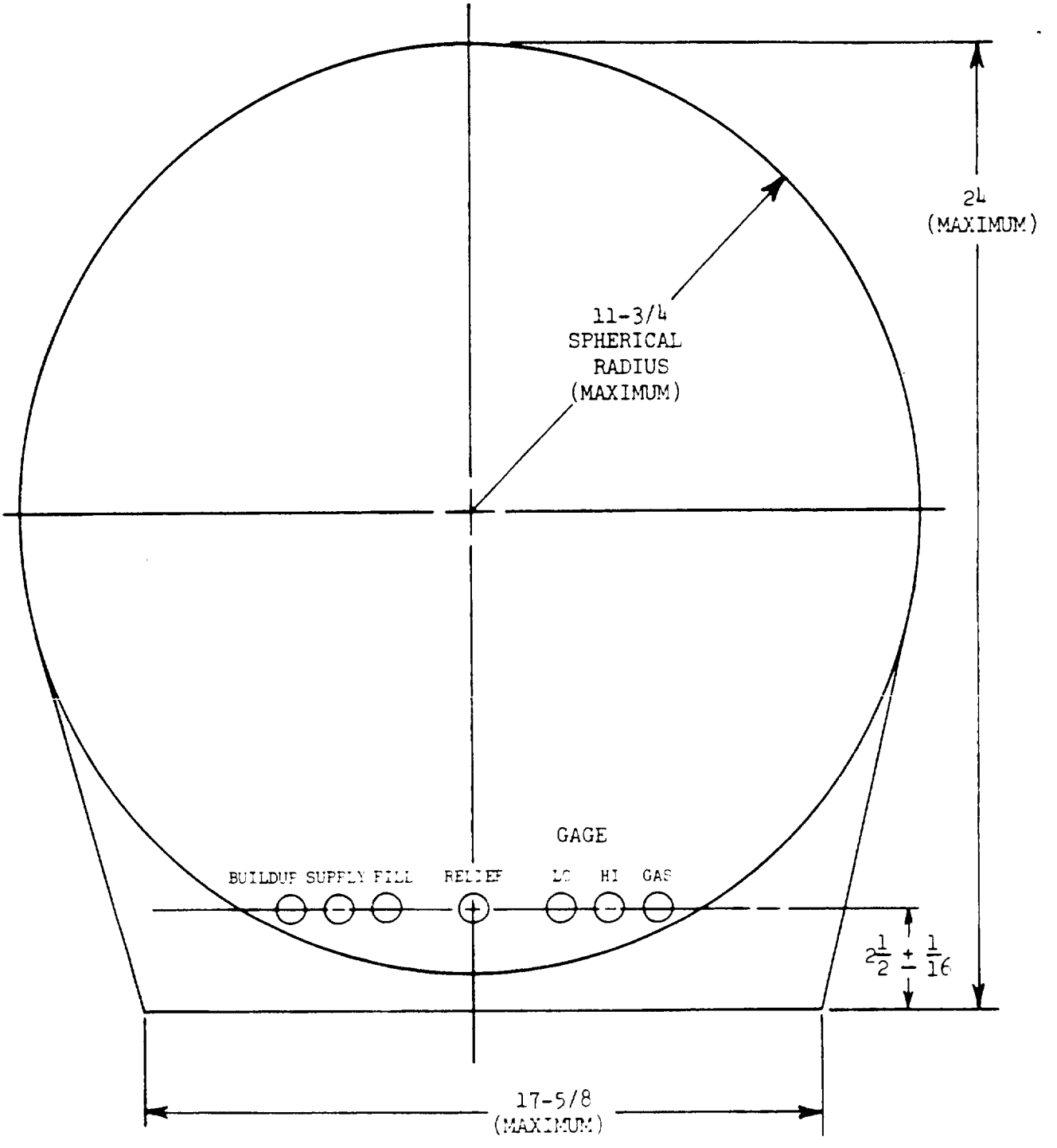
3.4.4 Buildup time. The time required for the converter to reach its operating pressure immediately after it has been filled shall not exceed 15 minutes.

3.4.5 Capacitance. The capacitance of the sensing element, after it is installed in the empty converter, shall be 910.0 (plus or minus 7.5) picofarad (pF).

3.4.6 Permanent volumetric expansion. The permanent volumetric expansion of the inner vessel shall not exceed 10 percent of the total volumetric expansion while the inner vessel is subjected to a pneumatic or hydrostatic expansion pressure of 500 psig.

3.4.7 Delivery rate. After the first 15 minutes and with the operating pressure remaining between 295 and 315 psig, the converter shall deliver constant gas flows up to 400 liters of gaseous oxygen per minute for not less than 2-1/4 hours. An intermittent increase or decrease of pressure by an additional 5 psig is allowable, but these variations shall not exceed a 2-minute interval for any one pressure cycle.

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DIMENSIONS IN INCHES

TOLERANCES:
FRACTIONS ± 1/64

FIGURE 1. Front view.

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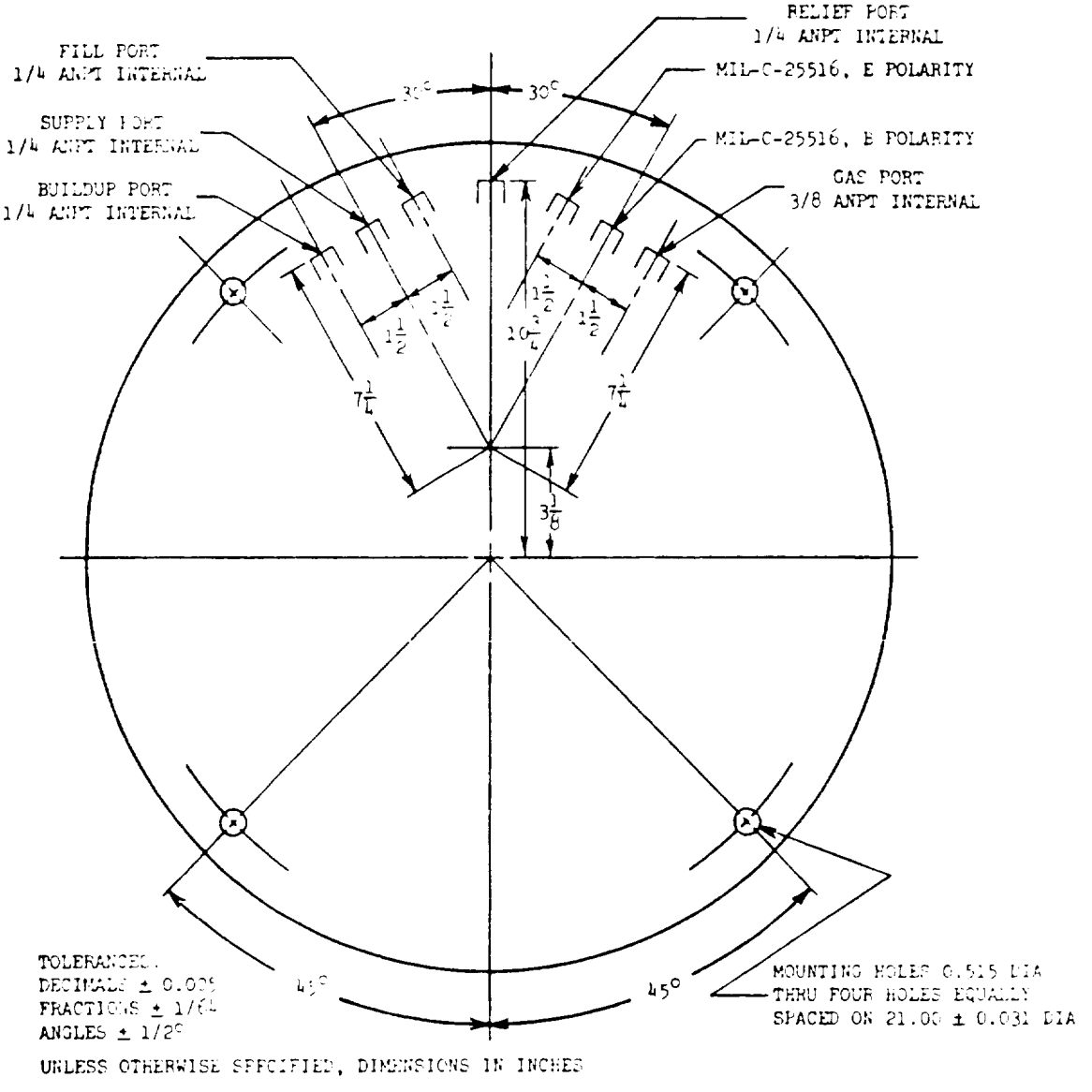


FIGURE 2. Bottom view.

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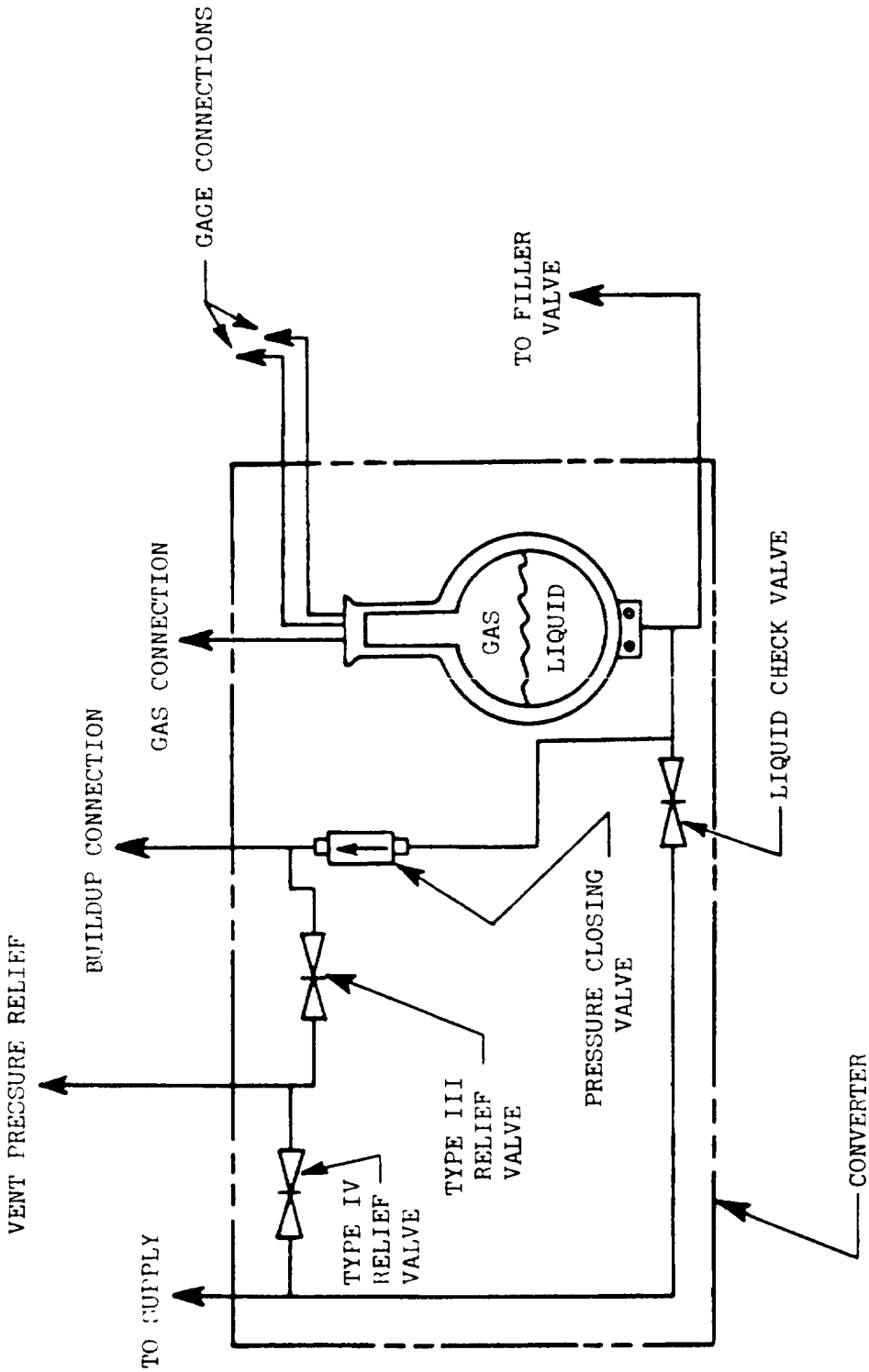


FIGURE 3. Schematic diagram of converter with capacitance gaging element.

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3.4.8 Calibration. The capacitance sensing element shall have characteristics that provide for a linear relationship between electrical capacitance and volume of liquid oxygen sensed. The relationship shall be in accordance with figure 4, using the equations that are specified in 3.4.8.1 to determine tolerances. An example of the use of these equations is as follows: If the actual quantity of liquid oxygen at the 30-liter test point is 30.2 liters, the measured capacitance must be between C maximum and C minimum.

C minimum = 5.725 X 30.2 + 900.6 = 1073.5 pF
C maximum = 5.875 X 30.2 + 919.4 = 1096.8 pF

3.4.8.1 Equations. The following equations shall be used to calculate the maximum and the minimum capacitances.

Values less than 15 liters	Values greater than 15 liters
C minimum = 5.6L + 902.5	C minimum = 5.725L + 900.6
C maximum = 6L + 917.5	C maximum = 5.875L + 919.4

C = Capacitance in pF
L = Exact quantity in liters of liquid at test point

3.4.9 Operating temperature. The operating pressure shall remain between 295 and 320 psig after the first 15 minutes of flow at the low temperature and shall remain between 290 and 315 psig after the first 15 minutes of flow at the high temperature. An intermittent increase or decrease of pressure by an additional 5 psig is allowable provided these variances do not exceed a 2-minute interval for any one pressure cycle.

3.4.10 Vibration. After the converter has been vibrated in accordance with MIL-C-25666, the evaporation loss of the converter in the buildup condition shall not exceed 5 pounds of oxygen.

3.5 Pressure relief valves. The pressure relief valves shall conform to types III and IV of MIL-V-9050 and shall be of such size and shape to fit the over-all dimensions specified for the converter.

3.6 Dimensions. The dimensions of the converter shall conform to figures 1 and 2.

3.7 Weight. The weight of the converter shall not exceed 80 pounds.

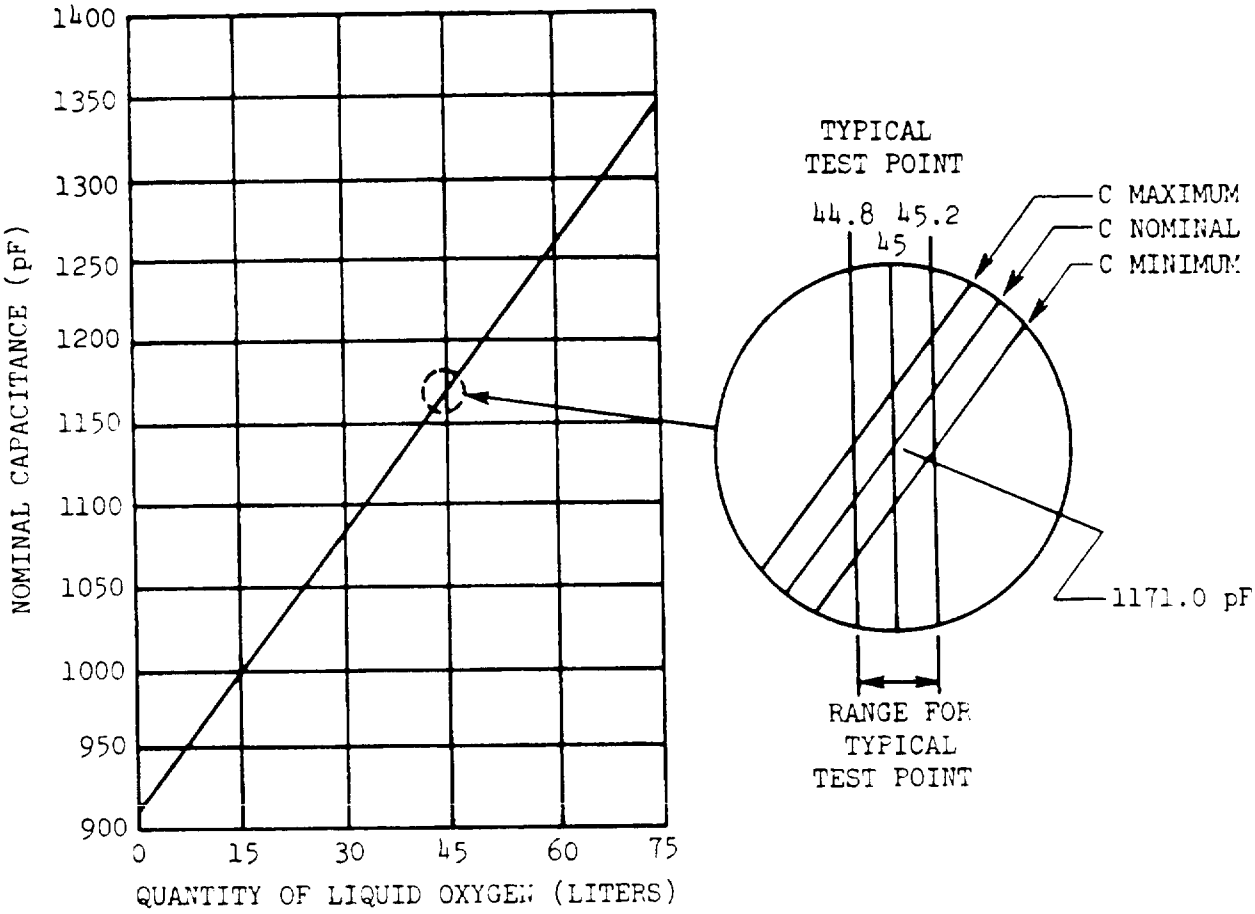
4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Classification of inspection. The inspection required by this specification is classified as follows:

(a) Quality conformance inspection
(b) Qualification inspection.

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LIQUID OXYGEN (LITERS)	NOMINAL CAPACITANCE (pF)
0	910.0
15	997.0
30	1084.0
45	1171.0
60	1258.0
75	1345.0

FIGURE 4. Relationship between capacitance and volume of oxygen sensed.

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4.3 Quality conformance inspection. The quality conformance inspection shall be in accordance with MIL-C-25666.

4.4 Qualification inspection. The qualification inspection shall consist of all of the methods of inspections that are described in MIL-C-25666.

4.4.1 Samples. At least two converters shall be subjected to the qualification inspection. One converter shall be submitted to the qualifying activity (see 6.3). The converter that is submitted to the qualifying activity may be a tested converter provided it has not been subjected to the gunfire test that is described in MIL-C-25666.

4.4.2 Report. Three copies of a qualification inspection report that has been prepared in accordance with MIL-STD-831 shall be submitted to the qualifying activity (see 6.3).

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with MIL-C-25666 (see 6.2).

6. NOTES

6.1 Intended use. The converter covered by this specification is intended to be used in the C-5, C-141, E-3 and E-4 aircraft to provide storage and pressurization of 75 liters of liquid oxygen to 300 psig nominal pressure.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Selection of applicable levels of preservation and packaging and packing of MIL-C-25666 (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they proposed to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for this Qualified Products List is the Aeronautical Systems Division (ASD/ENECE), Wright-Patterson Air Force Base, Ohio 45433, and information pertaining to qualification of products may be obtained from that activity.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Air Force - 11

Reviewer:
Air Force - 99

Preparing activity:
Air Force - 11

Project No. 1660-F412

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